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The Impact of Financial Literacy on Debt Levels

In recent years, markets have seen a rapid growth in the availability of financial products because technological advances have made these products easier to access for a wider range of consumers. Economists should be excited and encouraged to see such a development; recent literature on financial literacy, however, ought to mitigate excitement in the field. In a paper on consumer financial protection, Campbell, Jackson, Madrian, and Tufano (2011) site a 2004 Health and Retirement Survey study in which respondents were asked 3 basic questions on risk, the effect of inflation on interest rates, and accumulating interest. Only 24% of older respondents answered all three questions correctly. The younger respondents also did poorly with only 27% getting all three correct. These findings introduce an area of concern; many fiscal decisions, such as taking out a mortgage to purchase a home, are done with relative infrequency because of the magnitude of the purchase or the magnitude of the loan taken out. This leaves very little margin for error in situations involving large financial decisions and prevent many consumers from learning from their mistakes, saddling them with onerous financial responsibilities and debt that can be too much to handle.

In this paper, I will first address the issue of financial literacy and what the Consumer Financial Protection Bureau is doing to inform consumers. I will then look at existing literature on financial literacy, financial decisions, and present biased preferences. Next, I provide a statistical summary and brief data analysis of 2006 Fannie Mae mortgage acquisition data to provide rationale for further research on the importance of helping consumers making informed financial decisions. Finally, I will conclude by proposing research into the effects of financial literacy courses on debt levels.

Financial Literacy

Financial literacy can be defined as the "ability to process economic information and make informed decisions about financial planning, wealth accumulation, debt, and pensions" (Consumer Financial Protection Bureau). The importance of this skill, particularly as technology advances and as more financial products become available, cannot be understated. These new products can leave many consumers confused and unsure as to what they have bought. This also allows some in the financial industry to take advantage of the financially illiterate. The Consumer Financial Protection Bureau (CFPB) was founded in large part to combat both of these issues. They oversee much of the regulation of financial products that are put to market and give consumers tools to better understand the financial products available to them. Such tools include learning materials on auto loans, mortgages, and other financial products. Perhaps most interestingly, however, the CFPB offers a strategy for teaching financial literacy to students.

Their strategy largely focuses on what the CFPB calls "the three building blocks:" Executive Function (self-control and problem solving); Financial Habits and Norms (making spending and savings plans); and Financial Knowledge and Decision-making Skills (making informed decisions through research). They suggest that focusing on executive function at an early age is important because "it supports future orientation, perseverance, and planning abilities" (CFPB). Financial habits and norms are then introduced during middle childhood because the child will be able to plan and spend with moderation through observation (CFPB). Finally, the CFPB suggests that adolescents be exposed to financial education experiences that expands their knowledge, ability to research products, and analysis skills (CFPB). I will come back to this when I suggest research on a financial education course later in the paper.

Existing Literature

When judging the ability of households to make financial decisions, it is of integral importance that we look at homeownership so we may better understand the decisions being made; authors Goodman and Mayer find that "[t]he fact that homeownership is prevalent in almost all countries… suggests that most households still view homeownership as a critical part of a life-cycle plan for savings and retirement" (2018). In their paper, they conduct an analysis of homeownership rates by demographics before moving on to the returns to homeownership itself. The authors assert that homeownership, even after the recession, remains a valuable institution in the United States that allows households to build savings and accumulate wealth; they say that it is the "principal source of savings for most American households, especially households in the bottom part of the income distribution" (Goodman and Mayer 2018). Goodman and Mayer go on to challenge policy makers to remove these obstacles to homeownership rather than "pushing people to become homeowners for whom it doesn't make sense" (Goodman and Mayer 2018). In the context of financial literacy, it is therefore of integral importance that households understand the conditions upon which they sign a mortgage.

Much of the other literature reflects on the effects of financial illiteracy on preferences and economic decisions. Meier and Springer (2010), while not referencing financial literacy explicitly, discuss present bias and its impact on credit card debt. They find that "[c]onditional on borrowing, present-biased individuals borrow 25 percent more than dynamically consistent individuals." The authors also note that the correlation between present bias and credit card debt does not distinguish between those who are aware and those who are unaware of their present bias. Those who are aware of their own present bias are likely making informed decisions. While the authors acknowledge that

their results are not causal, they encourage further research to look for a causal link between present bias and borrowing because of the high correlation between the two variables in their results.

Robert H. Scott III (2007) notes that as of 2007, "[r]evolving credit card debt [was] close to $900 billion… The average United States household has eight credit cards, which are used to charge nearly $2 trillion in goods and services annually" (Scott 2007). He goes on to reflect on the history of credit cards as well as the failings of neoclassical economic theory to predict the volume of credit card usage we are seeing today. He instead prescribes Thorstein Veblen's social institutional theory of consumption; Veblen says that spending money is ingrained into a household and that "propensity for emulation… is a pervading trait of human nature (109)" (Veblen [1899], 1994). Essentially, Scott is using Veblen's theory to diagnose the problem of overborrowing and credit card use. Consumers want to be emulate those of the socio-economic class above them and will buy the most expensive items available to them to do so. He claims that the advent of higher liquidity in the form of credit cards and reduced regulatory discretion have created an atmosphere ripe for credit card abuse (Scott 2007). Taken together, these two papers paint a picture of a world with high liquidity and very little to prevent uniformed, present-biased individuals from being exploited by credit card companies.

Scott also offers a few policy options to address this issue, they are all on the supply side (i.e. stricter lending standards, making Chapter 7 bankruptcy easier to file, etc.) (2007). Many other academic papers focus on strengthening regulatory policy as well. For instance, authors Campbell, Jackson, Madrian, and Tufano worry that "[s]ome consumers may lack the cognitive capacity to optimize their financial situation" (2011). They argue that these limitations are often compounded by an inability to learn from mistakes; many financial transactions like purchasing a home are undertaken very few times in a consumer's lifetime (Campbell et al. 2011). The authors then use three case studies in mortgage choice, payday lending, and retirement saving to make the case for government

intervention and conclude by offering advice on the correct course of action regarding regulation of financial products (Campbell et al. 2011). These policies could very likely yield positive results, but they also bend to the issue at hand; financial illiteracy. These authors are merely responding to a symptom of financial illiteracy rather than correcting the initial issue. We can write regulatory policy until our wrists become sore and our fingers fall off, but each additional regulation will only decrease in its marginal effectiveness. At some point we become less like benevolent regulators and more akin to burdensome parents; consumers have an obligation to understand what they are purchasing. The longer we enable individuals to purchase financial products without properly understanding them, the worse the problem will become.

Credit cards and overborrowing, however, are only symptomatic of a larger problem. Lusardi and Mitchell (2014) perform a comprehensive review of the existing body of research on financial literacy. They begin by pointing out that financial products are increasingly complex while investors are often too unsophisticated to understand the simplest of products. The authors also complain of the constraints of current microeconomic models, saying that these models "assume that individuals execute saving and spend-down plans, which requires them to have the capacity to undertake complex economic calculations… however, few people seem to have much financial knowledge. Moreover, acquiring such knowledge is likely to come at a cost" (Lusardi and Mitchell 2014). They refer to their own paper written with another author, Michaud, prior to this publication that "calibrates and simulates a multiperiod dynamic life cycle model where individuals not only select capital market investments, but also undertake investments in financial knowledge" (Lusardi and Mitchell 2014).

Lusardi and Mitchell then reflect on US data and international data to find that financial illiteracy is quite widespread; respondents in the US were asked questions to measure their numeracy and capacity to do calculations related to interest rates, understanding of inflation, and understanding

of risk diversification. These same questions were then asked in similar national surveys in other countries. The results were rather discouraging; only two countries, Germany and Switzerland, had more than 50% of respondents get all three questions correct. The authors then go on to address financial literacy programs. They admonish the lack of literature on the effectiveness of financial literacy programs, and also note that because one cannot expect 100% participation in programs it is important to target specific groups of the population to account for various preferences and economic circumstances (Lusardi and Mitchell 2014).

A financial literacy course would therefore offer a novel way of expanding the knowledge of consumers without limiting their choices of products. It is a policy intervention that would allow regulators to work under the assumption of basic knowledge of the consumer, perhaps allowing them to take a more relaxed approach and freeing the market from any impediments in place that had been put there to allow for consumers' financial illiteracy. Before moving onto my proposed model, I offer a brief statistical summary and data analysis to reflect on the demonstrable need for financial literacy intervention.

Statistical Summary

I used Fannie Mae mortgage acquisition data, state unemployment data from the Bureau of Labor Statistics, and historical Federal Reserve interest rate benchmark data from the St. Louis Fed website to compile a data set for the years 2006 and 2014. I chose 2006 because we saw a great deal of market activity in the leadup to the recession; I chose 2014 because the recovery from the recession was under way. The economy had been overheating in 2006 and still was recovering during 2014, so each year represents an important one to the housing market and the economy as a whole. The Fannie Mae data is compiled of mortgages acquired by Fannie Mae in either 2006 or 2014. Below, I provide a

statistical summary of the dataset. I was forced to randomly cut the number of observations included from financial quarters 2, 3, and 4 of 2014 because the files were too large.

Table 1 - Summary Statistics across Dataset		
	2006	**2014**
Mortgage Interest Rate	6.451	4.326
Debt to Income Ratio	37.921	33.663
Borrower Credit Score	732.039	752
First Time Home Buyer	0.105	0.21
West Region	0.195	0.292
Northeast Region	0.157	0.124
Southeast Region	0.304	0.233
Southwest Region	0.106	0.142
Midwest Region	0.235	0.205
Observations	915,383	902,323
Average Unemployment	4.62%	6.25%

**Note: Mortgage Interest Rates and Debt to Income Ratio are shown in percentages while First Time Home Buyer and the Region variables are shown in ratios.*

I also perform basic OLS regression analyses using the dataset from above; in the first, I regress the debt-to-income ratio on credit score and interest rate. In the second, I regress the interest rate on the debt-to-income ratio and credit score. I controlled for first time homebuyers, state unemployment, the federal funds rate, region, and the channel through which the mortgage originated in both models. Much of the table falls in line with expectations (i.e. higher credit score is associated with lower interest rate and debt-to-income ratios across the board). It is interesting, however, to look at the coefficient on interest rate from the first model; in 2014, a one percentage point increase in the debt-to-income ratio was associated with a .0071% increase in the interest rate on one's mortgage. In contrast, a one percentage point increase in the interest rate was associated with a 2.56% increase in the debt-to-income ratio for the same year. In this comparison, the latter has far more economic

significance than the former and yields worrying results for those concerned with asymmetric information. While I recognize this argument may sound circumstantial, it must be taken with the existing literature. Prior research has shown that uninformed consumers abound; they may accept a higher interest rate on a mortgage without researching other offers, thereby locking themselves into higher payments. Therefore, this concern is certainly justified and provides reasonable grounds for implementing the strategy outlined in the next section.

Table 2 - Estimates of OLS Coefficients						
	2006		**2014**		**2006 and 2014**	
	Debt-to-Income	**Interest Rate**	**Debt-to-Income**	**Interest Rate**	**Debt-to-Income**	**Interest Rate**
Credit Score	-0.02598 [.00022]	-0.00126 [6.40X10^-6]	-0.02754 [.0002]	-0.00259 [.0000104]	-0.0267 [.00015]	-0.00182 [5.87X10^-6]
Debt-to-Income	n/a	0.00128 [0.0000305]	n/a	0.0071 [.0000551]	n/a	0.00332 [.0000291]
Interest Rate	1.506 [.0359]	n/a	2.558 [0.0199]	n/a	2.153 [.0189]	n/a
Observations	910,002		898,091		1,808,093	

**Note: The estimates shown control for state unemployment, the federal funds rate, region, and channel of origination. Standard errors are shown in brackets. All coefficients shown are significant at the 1% level.*

Expanding Financial Literacy

I would like to investigate the effects of financial literacy courses on spending habits and financial decisions, particularly on the amount of debt consumers take on. Before continuing, I must address the issue of debt. With this paper, I neither condemn nor condone its use—I am merely trying to ascertain the effects of financial literacy on debt levels. There are healthy and unhealthy levels of debt and those levels change from household to household based on their preferences. The importance of recognizing future wealth or the lack thereof, however, is my principle concern. The purpose of educating households on financial literacy is primarily to get them to better understand the benefits and costs of their transactions in the present and the effects of those transactions on the future. Many

financially illiterate households leave themselves vulnerable to the abuses of pay day lenders and credit card companies because they are unable to properly identify the costs of high interest loans on future wealth (Lusardi and Mitchell 2014).

Thankfully, the Consumer Financial Protection Bureau (CFPB) has done a great deal of research into the most effective techniques for teaching consumers. They also offer a curriculum review tool that allows educators to judge the effectiveness of their own courses, providing a prospective path for standardization across schools and communities.

The CFPB emphasizes exposure through an individual's development; that said, I would like to select schools at random in which I would integrate financial literacy education into the curriculum. This curriculum would ideally follow these students from grade school through high school, teaching the building blocks and following the timeline listed earlier in the paper. It would also be introduced across grade levels in order to control for age in future studies. The students given the treatment would be compared with other schools in the community that were not given the treatment. I would like to "treat" the entire school to the new curriculum rather than a certain number of students to avoid spillover effects; it could be reasonably assumed that the students would share information between each other, particularly as they got older.

I would also think that the Chicago Public School (CPS) system, or a school system in another metropolitan area, would present the ideal sample for this given study for several reasons. For one, the sample size of students would likely be sufficiently large for the study's findings to be significant. Furthermore, the high density of students would provide a large concentration of schools in a given area. I would then be able to control for income, property value, and other variables associated with people living near one another. This high density comes at a price. It would be likely that friends and family would be spread across school districts—this raises the concern of spillover effects between

said family and friends. Given that these interactions take place outside of school, however, it is reasonable to assume that they would be muted. Siblings and relatives within a home would present the largest concern of spillover effects; it's more unlikely that the material would be shared with extended family and friends beyond the treated school's boundaries.

Then I would contact the students from the CPS, across five grade levels, every five years to compile panel data on their financial decisions over time as well as the depreciation of their financial literacy. This would allow me to control for educational attainment beyond high school, marital status, and other variables that would likely affect financial literacy. Surveying students across grade levels would allow me to control for differences between grade levels. I would measure the depreciation in financial literacy by asking questions regarding financial decisions that would test the same abilities and would be of similar difficulty. It is also important to note that school districts nationwide vary in quality, funding, amongst other characteristics. Compiling such comprehensive panel data would allow controls for time invariant characteristics of a given school district.

Measuring the retainment of financial literacy is instrumental to the study, so I would include different questions of similar difficulty on each survey. Most impactful economic decisions, such as buying a house or making other large investments, are not made immediately after high school; if the students are unable to retain the lessons learned regarding financial literacy by the time they need to make these decisions, then the efficacy of the new curriculum would likely be questionable at best. Similarly, it would be interesting to see how their feelings about debt change and how much debt the two groups of students accumulate. I'd also like to include questions that measure time preferences to see how these preferences change over time and how the curriculum has changed them. Present-bias, as shown in the papers above, is often correlated with high credit card debt and taking on debt with high interest rates. This high interest debt can prove detrimental to personal finance later in life.

With this very comprehensive data set, I would like to use a 2 stage least squares model using the financial literacy course as an instrumental variable. I would then get fitted values for financial literacy from the first stage:

$$financial\ literacy_r = \omega + \rho_1 fin\ lit\ course_r + \rho_2 home\ ownership_r + \rho_3 present\ bias_r + \rho_4 X_r + u_r.$$

Using these fitted values of financial literacy, I would test the following models separately:

$$low\ interest\ debt_r = \alpha + \beta_1 \widehat{financial\ literacy}_r + \beta_2 home\ ownership_r + \beta_3 present\ bias_r + \beta_4 X_r + u_r,$$

$$high\ interest\ debt_r = \epsilon + \delta_1 \widehat{financial\ literacy}_r + \delta_2 home\ ownership_r + \delta_3 present\ bias_r + \delta_4 X_r + u_r,$$

where r represents an index over individuals and where X_r is a vector of income, age, education level, and other individual characteristics. So long as the course were to have a measurable impact on an individual's financial literacy, it would certainly be relevant. Exogeneity is a little trickier. The course curriculum may bias consumers toward low interest debt because it is typically considered safer. This would effectively change their preferences for debt. However, I would argue that this change in consumption can be attributed to increased financial literacy rather than a change in preferences because the individual would be more aware of the dangers of high interest debt. Therefore, the financial literacy course instrumental variable would be uncorrelated with both the high interest and low interest debt variables except for its correlation with financial literacy, establishing exogeneity.

I've also included home ownership in the model because, as stated above in the paper regarding home ownership, purchasing a home is often the largest financial decision made by any household in their lifetime (Goodman and Mayer 2018). High interest debt would refer to credit card debt, pay day loans, and other sources of debt that are typically considered risky while low interest debt would refer to mortgages, auto loans, etc. I've discriminated between the two largely because the former would typically constrain the financial opportunities of households while the latter is typically considered healthy, or at least safer, debt. Additionally, the nature of the panel data collected would allow monitoring of how preferences and attitudes toward debt change over time. This allows us to

see how financial literacy retainment affects debt levels and, by extension, financial opportunities available to households.

Reflections

The introduction of the "Expanding Financial Literacy" section brings us to the crux of this paper and its relevance to the market as a whole: asymmetric information. Uninformed consumers, particularly the elderly, find themselves unwitting victims of fraud (Lusardi and Mitchell 2014). Even worse, many households find themselves saddled with high interest debt because they didn't recognize the costs of their transgressions in a prior period. The Consumer Financial Protection Bureau has been put in place to protect households against many of these scams. They are not, however, present to prevent each micro-aggression nationwide. Therefore, I would like to emphasize the importance of a mixed approach to financial protection of the consumer; regulation of the market and educating consumers offers households a means of protecting themselves while giving regulators an opportunity to assume basic financial knowledge on an individual level. After all, the CFPB cannot prevent someone from swiping their credit card even if prevention is in their best interest (so long as the purchase is lawful).

Judging the efficiency of the CFPB's building blocks of financial literacy is therefore the logical place to start. More regulation on the market will only yield marginal returns to consumer wellbeing, but if we can teach them to make informed spending decisions, much of the regulation will seem redundant.

Bibliography

"Archived News Releases | State Employment and Unemployment." *U.S. Bureau of Labor Statistics*, U.S. Bureau of Labor Statistics, 2018, www.bls.gov/bls/news-release/laus.htm.

"Effective Federal Funds Rate." *FRED*, 2 Apr. 2018, fred.stlouisfed.org/series/fedfunds.

Campbell, John Y., Jackson, Howell E., Madrian, Brigitte C., and Tufano, Peter. "Consumer Financial Protection." *The Journal of Economic Perspectives* 25, No. 1 (Winter 2011): 91-113.

Goodman, Laurie S., and Mayer, Christopher. "Homeownership and the American Dream." *The Journal of Economic Perspectives* 32, No. 1 (Winter 2018): 31-58.

Lusardi, Annamaria, and Mitchell, Olivia S. "The Economic Importance of Financial Literacy: Theory and Evidence." *Journal of Economic Literature* 52, 1 (March 2014): 5-44.

Meier, Stephan, and Charles Sprenger. "Present-Biased Preferences and Credit Card Borrowing." *American Economic Journal: Applied Economics* 2, No. 1 (January 2010): 193-210.

"Research & Reports." *Consumer Financial Protection Bureau*, www.consumerfinance.gov/data-research/research-reports/?topics=data.

Scott, Robert H. "Credit Card Use and Abuse: A Veblenian Analysis." *Journal of Economic Issues* 41, No. 2 (June 2007): 567-574.

Veblen, Thorstein. *The Theory of the Leisure Class.* Reprint, New York: Penguin Books, [1899] 1994.